

DHAHARAN KING ABDULAZIZ AB, Saudi ArabiaWMO: **404160**Lat: **26.265N**Lon: **50.152E**Elev: **84**StdP: **14.65**Time Zone: **3.00 (E03)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	46.5	49.0	24.4	18.6	65.2	28.4	22.4	69.5	23.4	62.4	21.8	62.4	9.6	300	0.509

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.8	114.7	73.2	112.6	73.3	109.8	73.5	88.3	96.6	86.7	96.1	85.0	95.4	13.1	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
86.4	193.3	93.7	84.4	181.3	93.0	82.5	169.8	92.8	54.2	95.8	51.7	95.4	49.8	95.0	97.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.4	21.8	19.7	42.1	118.2	2.9	2.1	40.1	119.7	38.4	121.0	36.8	122.2	34.7	123.7		
(5)			35.5	90.9	3.4	2.7	33.0	92.8	31.0	94.4	29.1	95.9	26.6	97.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	81.7	61.5	64.5	71.4	80.6	90.9	96.4	98.7	97.9	92.8	85.7	74.1	64.8
	DBStd	14.07	4.67	5.20	5.30	5.27	4.36	3.20	2.50	2.64	3.51	3.96	5.51	5.02
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	272	129	65	8	0	0	0	0	0	0	0	4	66
	CDD50	11573	356	407	663	917	1269	1393	1510	1485	1285	1106	723	459
	CDD65	6370	20	52	206	467	804	943	1045	1020	835	641	277	60
	CDH74	94353	64	241	1555	5183	12387	16025	17975	17186	12996	8230	2250	261
	CDH80	63512	3	36	536	2570	8184	11721	13512	12722	8793	4641	772	22
Wind	WSAvg	9.6	9.3	10.2	10.3	10.2	10.4	11.4	10.3	8.6	8.4	7.9	8.9	9.1
Precipitation	PrecAvg	3.0	0.6	0.5	0.7	0.3	0.0	0.0	0.0	0.0	0.0	0.0	0.4	0.5
	PrecMax	12.9	3.2	3.1	8.2	2.3	0.4	0.0	0.0	0.0	0.0	0.7	4.8	3.7
	PrecMin	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	2.6	0.7	0.8	1.4	0.4	0.1	0.0	0.0	0.0	0.0	0.1	0.9	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	79.1	84.3	96.6	104.0	113.0	116.7	116.9	116.4	112.6	105.8	95.1	82.6
		MCWB	60.4	63.0	64.8	68.5	71.3	71.2	73.8	73.9	73.2	70.3	70.2	65.9
	2%	DB	75.4	80.2	89.7	99.0	109.0	113.4	114.6	113.1	109.2	102.2	90.0	79.1
		MCWB	61.3	60.9	63.3	67.8	71.1	71.8	73.5	75.1	73.2	71.6	69.8	64.0
	5%	DB	73.1	76.9	85.8	95.4	105.8	111.1	112.2	111.0	106.5	99.0	87.7	77.0
		MCWB	60.8	60.8	62.7	66.6	70.0	71.4	73.2	75.7	73.8	71.9	69.4	63.8
	10%	DB	70.2	73.7	82.2	91.8	102.5	108.0	109.7	109.0	104.1	96.6	84.4	74.4
		MCWB	59.5	60.2	61.9	66.6	70.0	71.1	73.5	76.3	74.6	72.2	68.7	63.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	67.6	68.5	71.5	75.4	82.8	84.6	89.9	91.1	88.9	83.5	79.2	71.2
		MCDB	72.0	75.5	81.1	87.9	94.7	94.8	96.6	97.2	96.6	89.9	85.1	76.6
	2%	WB	65.4	66.2	69.1	72.9	79.2	81.7	87.3	89.2	86.1	81.3	75.5	68.9
		MCDB	70.4	72.1	78.2	86.2	93.1	93.5	96.7	96.9	94.5	88.1	83.1	74.4
	5%	WB	63.5	64.7	67.4	71.4	76.9	79.2	85.3	87.5	84.0	79.8	73.6	66.9
		MCDB	69.3	71.5	76.7	85.4	92.2	94.7	95.9	96.3	93.6	87.5	81.5	72.8
	10%	WB	61.6	63.1	65.7	69.9	74.7	76.9	83.2	86.1	82.1	78.5	71.8	65.0
		MCDB	68.1	70.3	76.1	84.6	92.3	95.9	95.4	96.0	92.5	87.1	80.1	71.6
Mean Daily Temperature Range	5% DB	MDBR	17.5	18.4	20.1	21.5	23.4	24.7	23.8	23.0	24.0	22.8	19.0	18.0
		MCDBR	19.2	21.0	25.9	26.6	27.7	28.6	27.2	26.2	26.5	26.2	21.9	20.0
		MCWBR	10.0	9.0	9.9	9.4	10.3	11.7	12.4	13.1	12.7	11.4	10.2	10.3
	5% WB	MCDBR	15.7	16.3	19.1	22.3	24.1	26.0	22.8	21.6	22.5	22.5	19.2	16.1
		MCWBR	10.2	9.0	9.5	9.6	11.3	13.3	12.5	11.7	12.3	11.3	10.2	10.0
Clear-Sky Solar Irradiance	taub	0.520	0.571	0.632	0.692	0.687	0.656	0.686	0.636	0.565	0.527	0.533	0.501	
	taud	1.839	1.731	1.606	1.541	1.539	1.582	1.570	1.670	1.790	1.870	1.856	1.896	
	Ebn at Noon	224	224	219	211	212	218	211	221	233	233	219	223	
	Edh at Noon	58	69	83	90	91	86	87	78	68	59	56	53	
All-Sky Solar Radiation	RadAvg	1182	1416	1697	1889	2174	2347	2221	2115	1950	1624	1220	1115	
	RadStd	79	89	121	116	64	65	50	40	37	47	85	80	